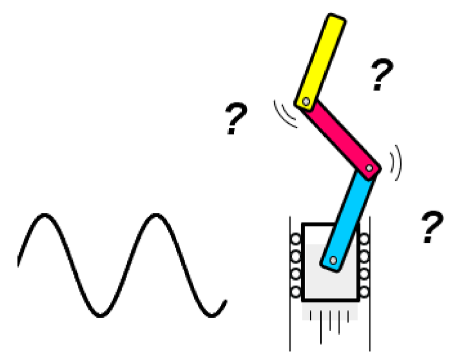

Balancing an Inverted Pendulum by Oscillating It

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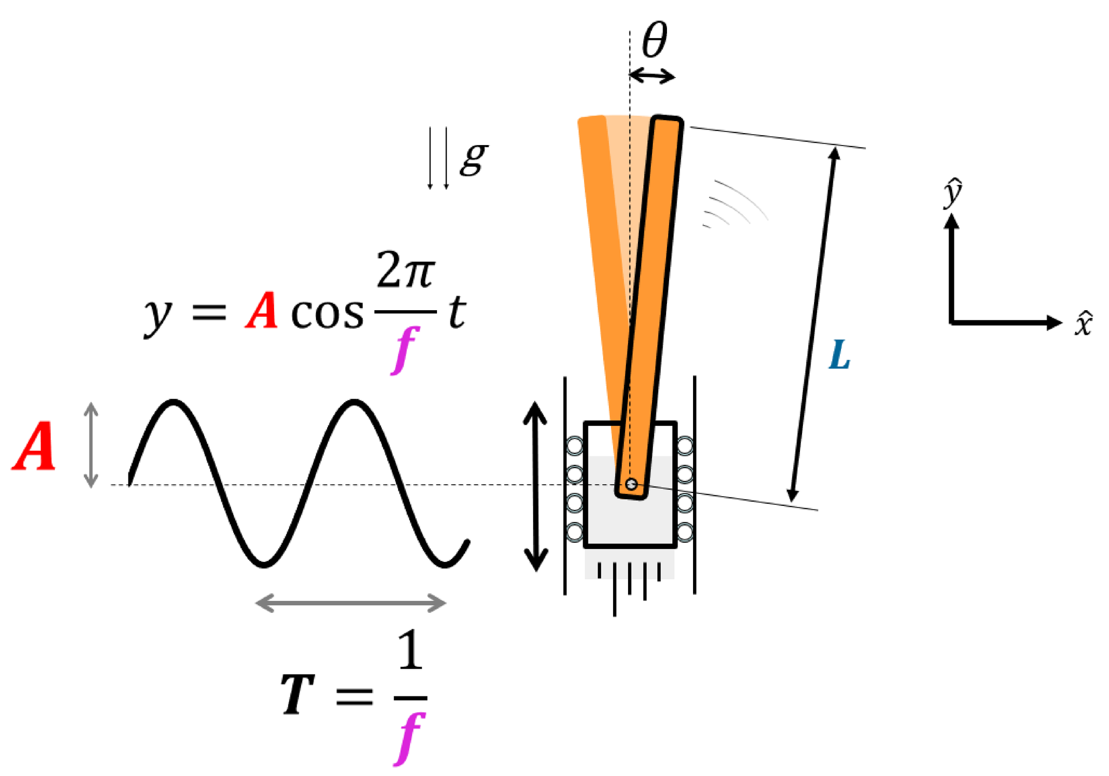
Abstract

Oscillating a pendulum rapidly up and down can, rather surprisingly, make it stay upright. This unusual behavior is known in literature as "Kapitza's Pendulum," and it is useful to the understanding of a class of oscillatory systems. In order to explore the behavior of such systems, a speaker setup was used to measure the stability of an inverted single and double pendulum. The single pendulum results are consistent with existing theory. Results for frequencies between 16 – 30 Hz yield stability criterion constants $\gamma_{local} = 0.142 \pm 0.005\ [m\ Hz]$ and $\gamma_{global} = 0.15 \pm 0.06\ [m\ Hz]$ being 3% and 2% different from the theoretical prediction, $\gamma_{theoretical} = 0.147\ [m\ Hz]$, respectively.

Theoretical Prediction for Single Pendulum System



Definition of stability criterion, γ

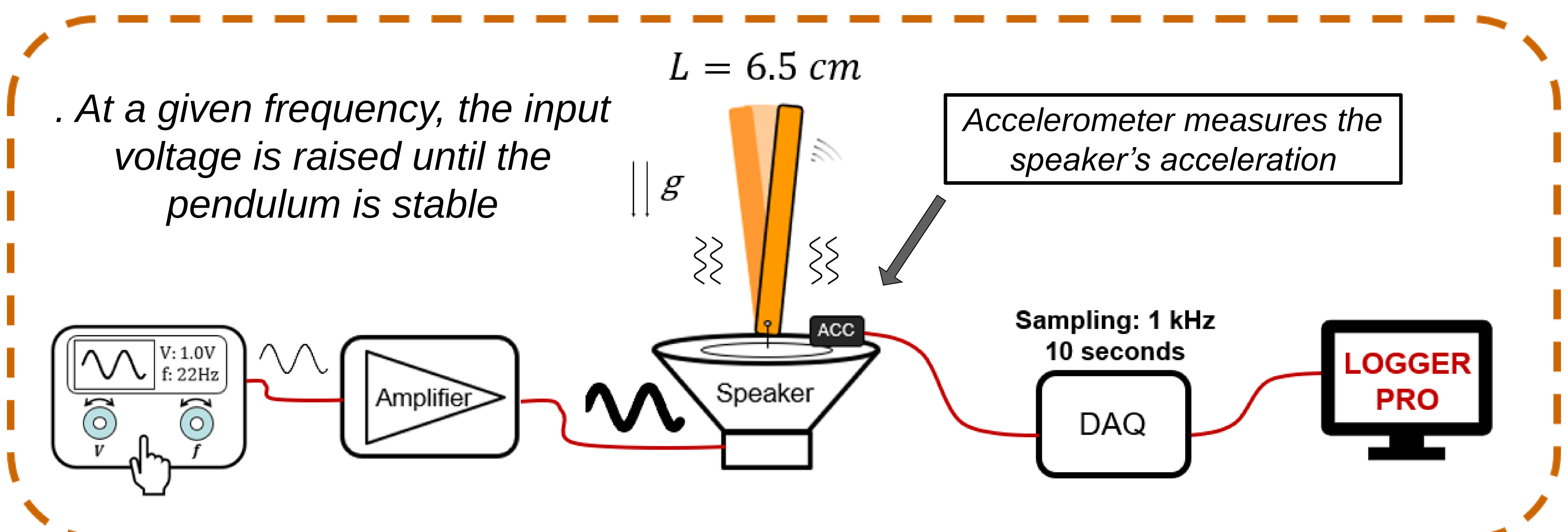
$$Af \geq \frac{1}{2\pi}\sqrt{\frac{4}{3}gL} \equiv \gamma$$

[1]

If $Af \geq \gamma$, the pendulum will be **stable in an upright position**

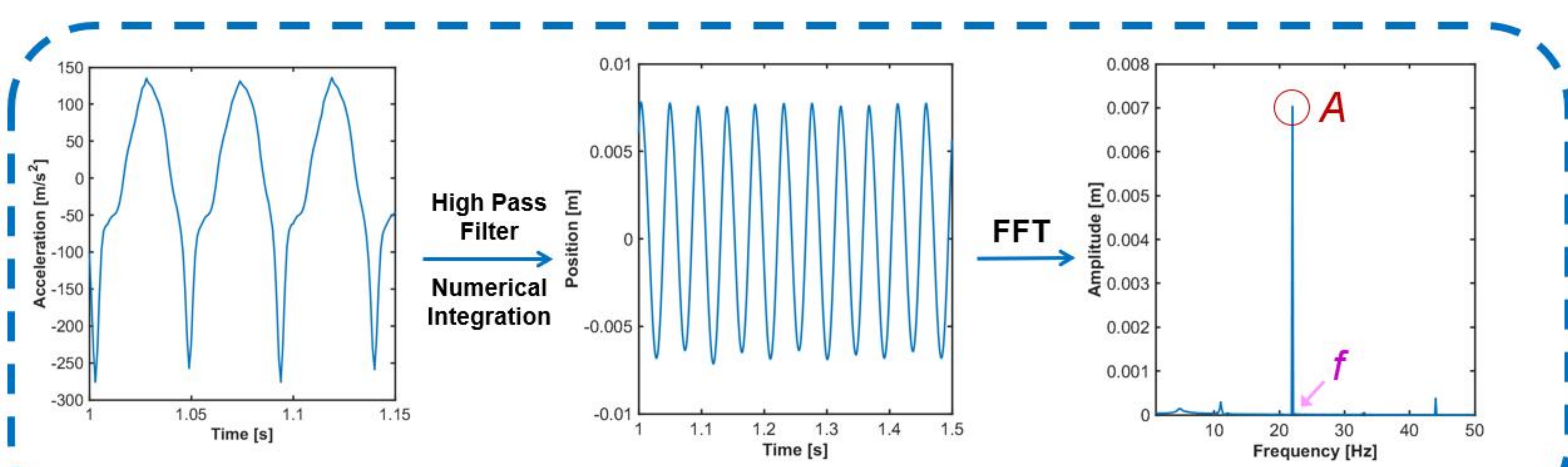
Data Acquisition & Analysis

Data Acquisition



Data are collected for frequencies between 16 and 30 Hz

Data Analysis

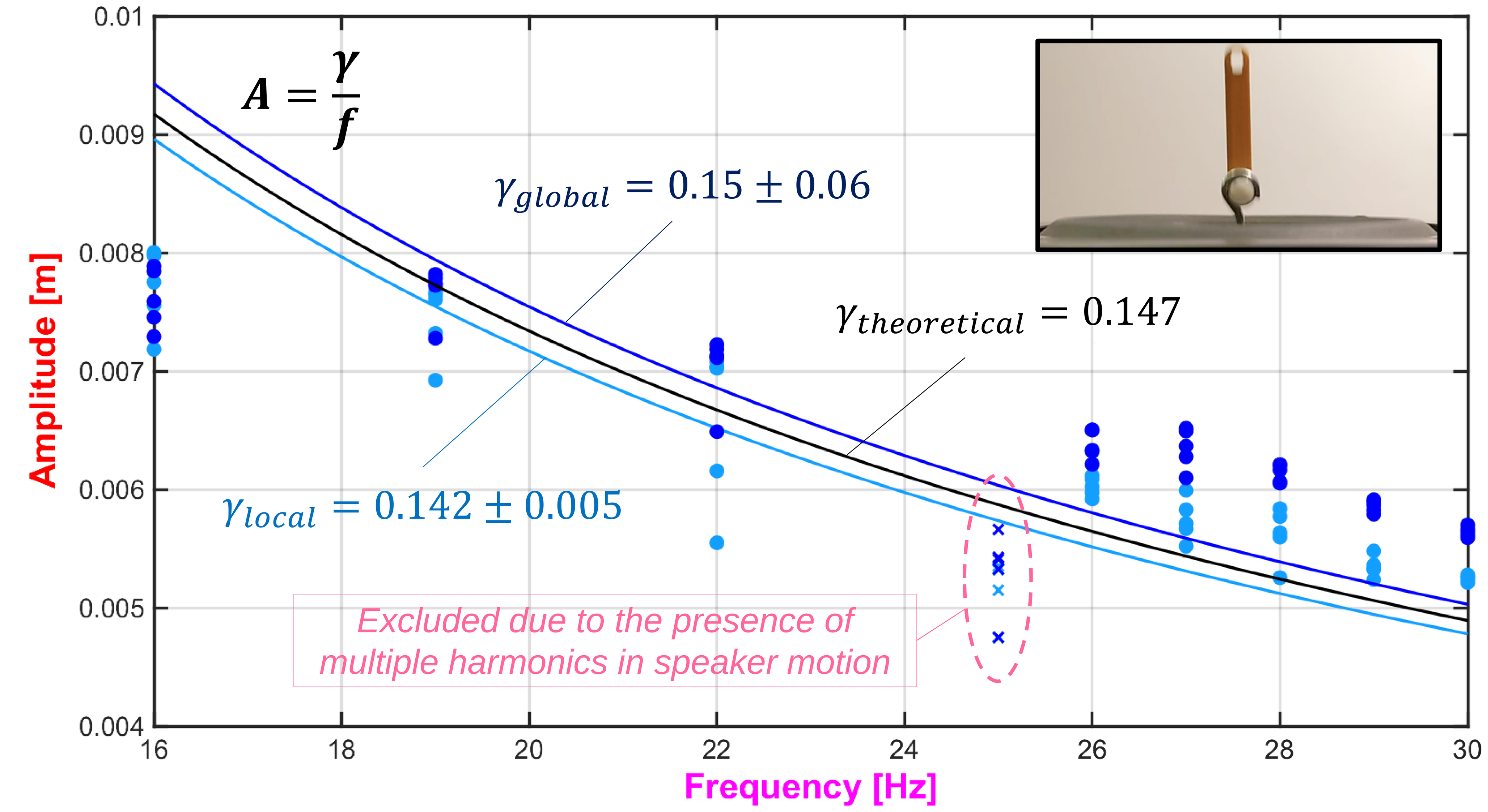


Raw Acceleration Data **Position of Pendulum** **Stable Combination**

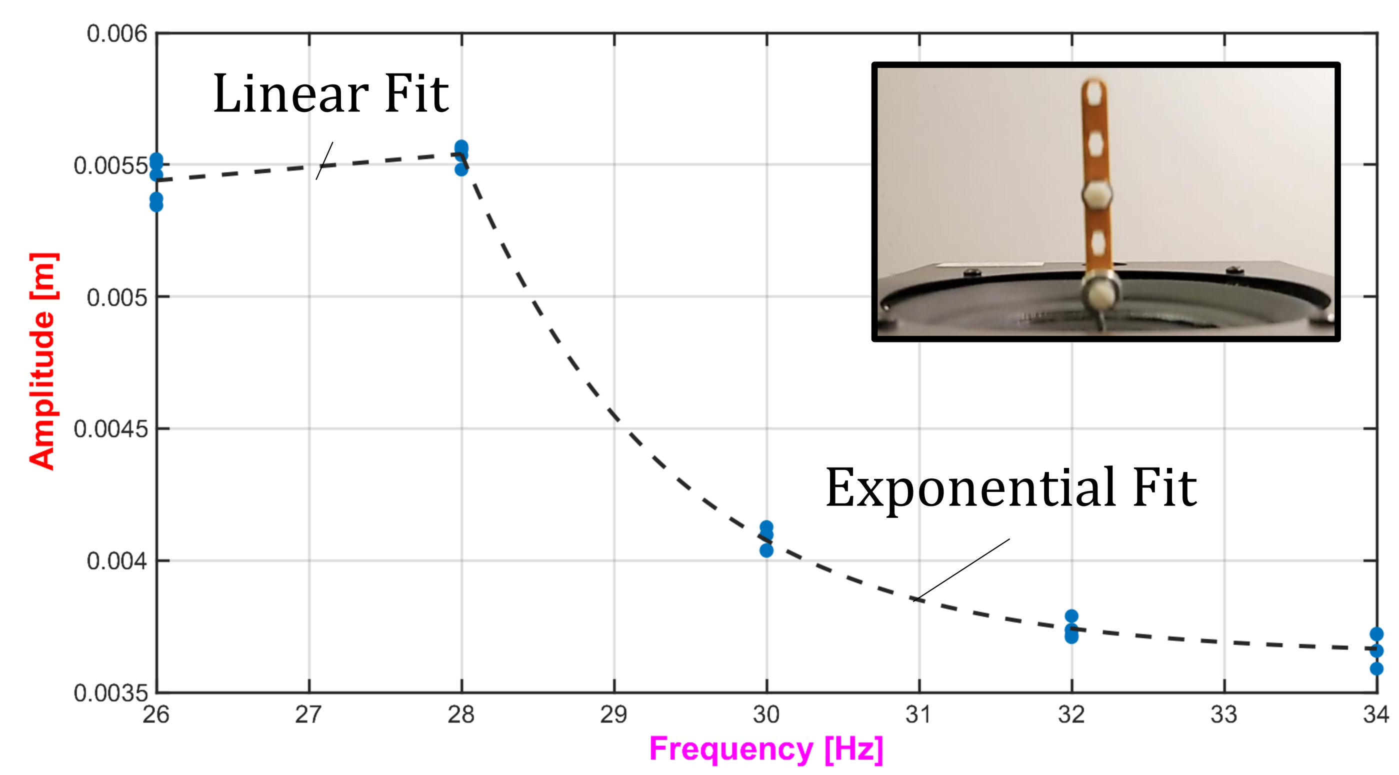
Stability Results for Single Pendulum System

Local Stability: Pendulum is stable after being raised to the vertical position.

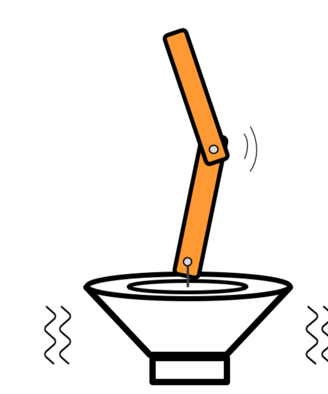
Global Stability: Pendulum gets to stable position without any assistance.



Addendum – Exploration of Double Pendulum System



Local Stability is also achieved for double pendulum system. Empirical fits are shown.



$L_1 = L_2 = 3\ cm$

Conclusion

- The experimental results for a single pendulum system show good agreement with the model, with $\gamma_{local} = 0.142 \pm 0.005\ [m\ Hz]$ and $\gamma_{global} = 0.15 \pm 0.06\ [m\ Hz]$ being 3% and 2% different from the theoretical prediction, respectively.
- Theoretical model prediction, $\gamma_{theoretical}$, lies in between the experimental fits of local stability & global stability results ($\gamma_{local} < \gamma_{theoretical} < \gamma_{global}$).
- Comparison of the measured behavior of the double pendulum system with theoretical prediction might explain the observed piecewise fits.

Acknowledgements

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References

[1] L. D. Landau and E. M. Lifschitz, Mechanics, Pergamon, New York (1976), pp. 93 – 95.

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